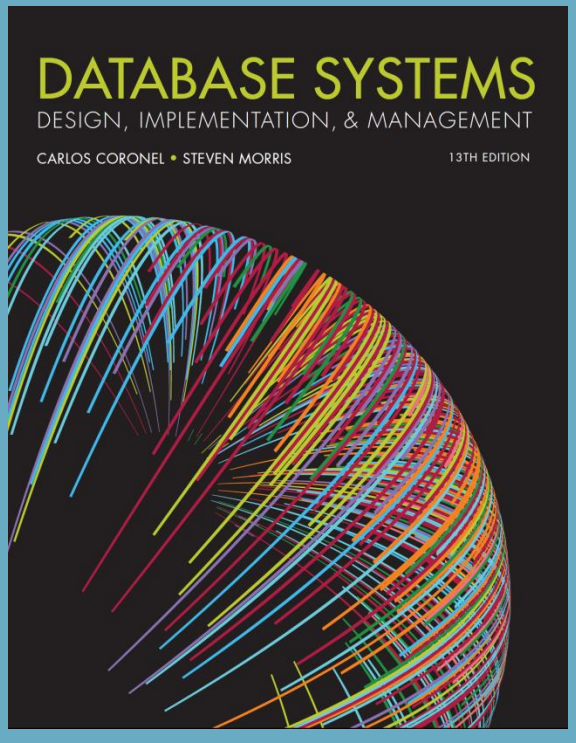


Chapter 4

Entity Relationship (ER) Modeling

Learning Objectives

- Chapter 4:
  - Identify the main characteristics of entity relationship components
  - Describe how relationships between entities are defined, refined, and incorporated into the database design process
  - See how ERD components affect database design and implementation
  - Understand that real-world database design often requires the reconciliation of conflicting goals
- Chapter 5:
  - The characteristics of good primary keys and how to select them
  - How to use flexible solutions for special data-modeling cases

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The Entity Relationship Model (ERM)

- Forms the basis of an entity relationship diagram (ERD)
  - Conceptual database as viewed by end user
- Database's main components
  - Entities
  - Attributes
  - Relationships

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Entities

- Object of interest to the end user
  - Refers to the entity set and not to a single entity occurrence
- ERM corresponds to a table—not to a row—in the relational environment
  - ERM refers to a table row as an entity instance or entity occurrence
- In Chen, Crow's Foot, and UML notations, an entity is represented by a rectangle that contains the entity's name
  - The entity name, a noun, is usually written in all capital letters

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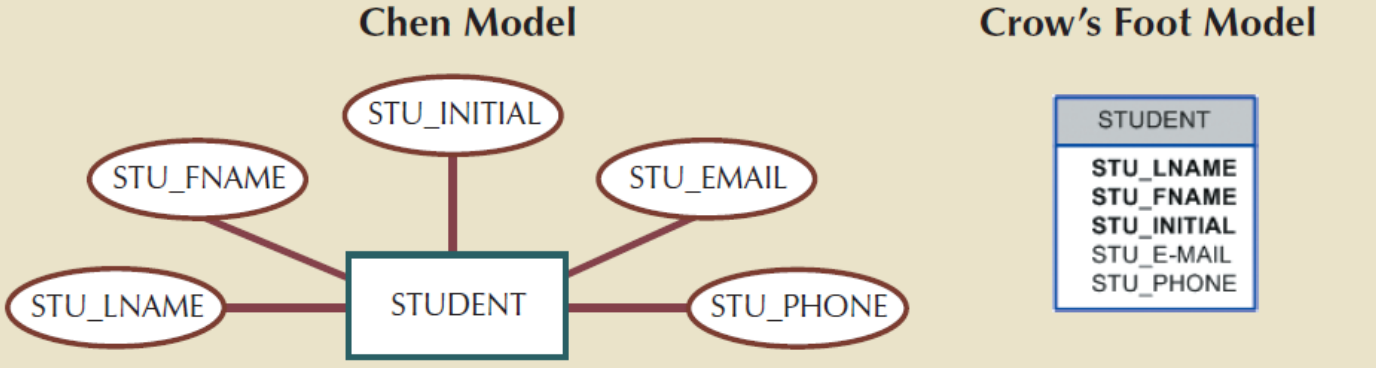
Attributes (1 of 7)

- Characteristics of entities
  - Required attribute: must have a value and cannot be left empty
  - Optional attribute: does not require a value and can be left empty
  - Domain: set of possible values for a given attribute
  - Identifier: one or more attributes that uniquely identify each entity instance
  - Composite identifier: primary key composed of more than one attribute
  - Composite attribute: attribute that can be subdivided to yield additional attributes
  - Simple attribute: attribute that cannot be subdivided
  - Single-valued attribute: attribute that has only a single value
  - Multivalued attributes: attributes that have many values

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Attributes (2 of 7)

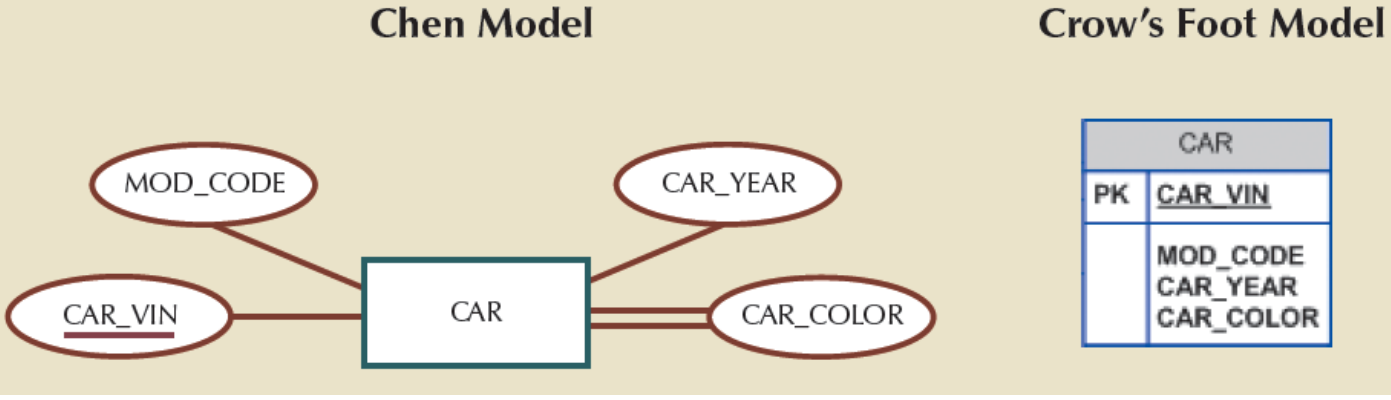
FIGURE 4.1 THE ATTRIBUTES OF THE STUDENT ENTITY: CHEN AND CROW' FOOT



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Attributes (3 of 7)

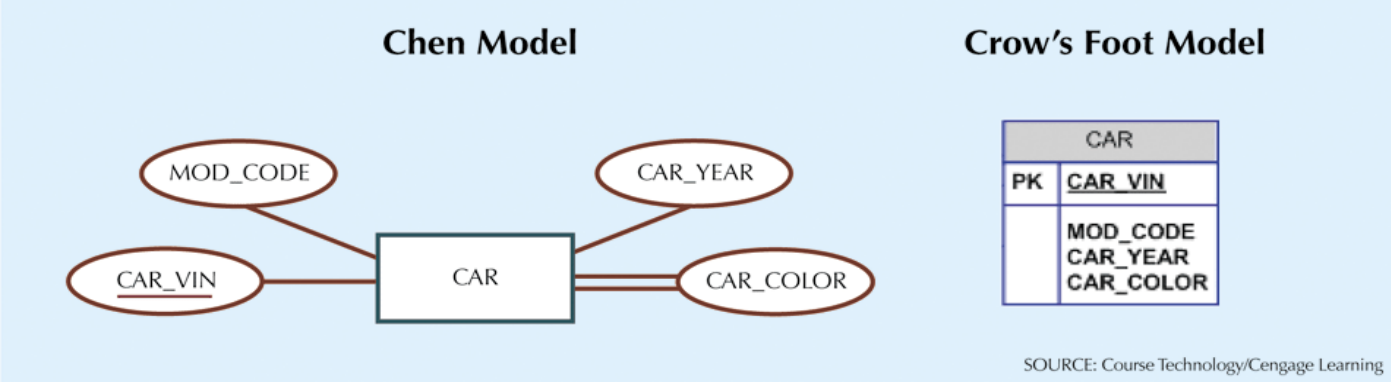
FIGURE 4.3 A MULTIVALUED ATTRIBUTE IN AN ENTITY



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Attributes (4 of 7)

- Requirements of multivalued attributes
  - Create several new attributes, one for each component of the original multivalued attribute
  - Develop a new entity composed of the original multivalued attribute's components



Car_COLOR is multivalued as it involves TopColor, TrimColor, BodyColor

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Attributes (5 of 7)

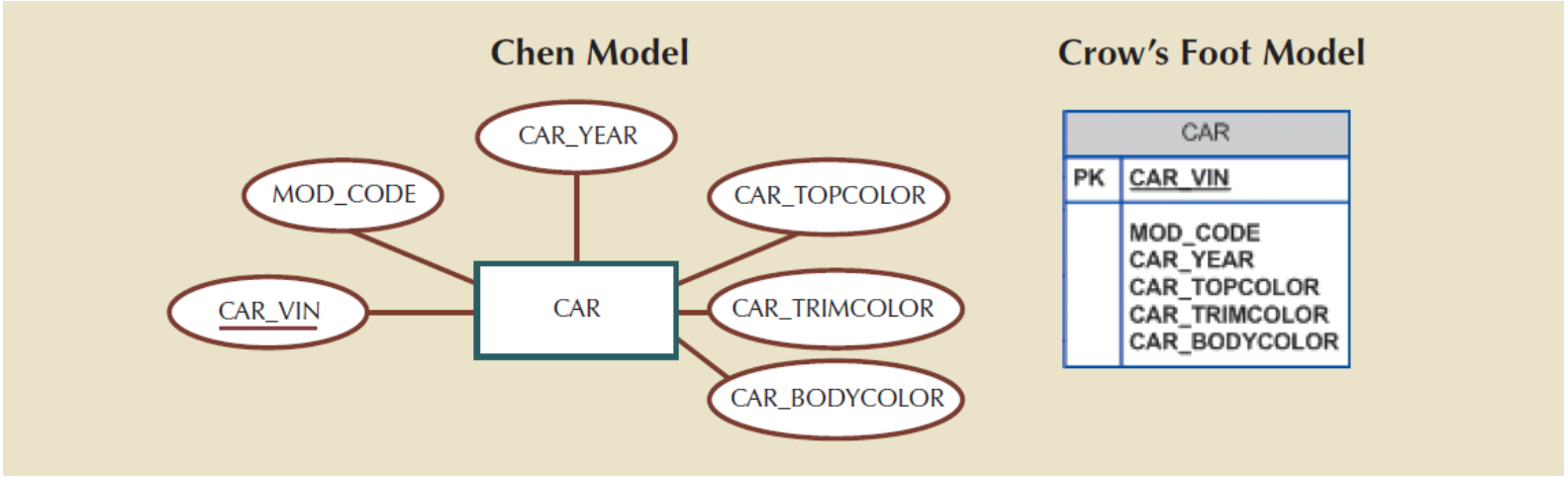
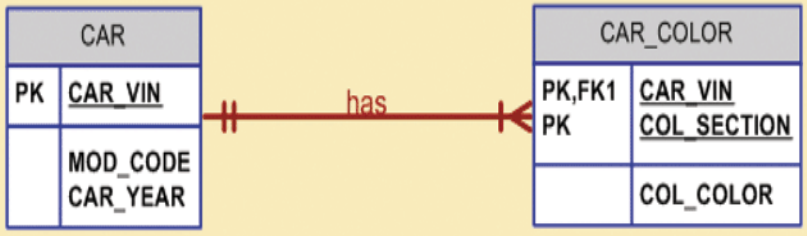


TABLE 4.1 Components of the Multivalued Attribute

SECTION	COLOR
Top	White
Body	Blue
Trim	Gold
Interior	Blue

FIGURE 4.5 A new entity set composed of a multivalued attribute's components



SOURCE: Course Technology/Cengage Learning

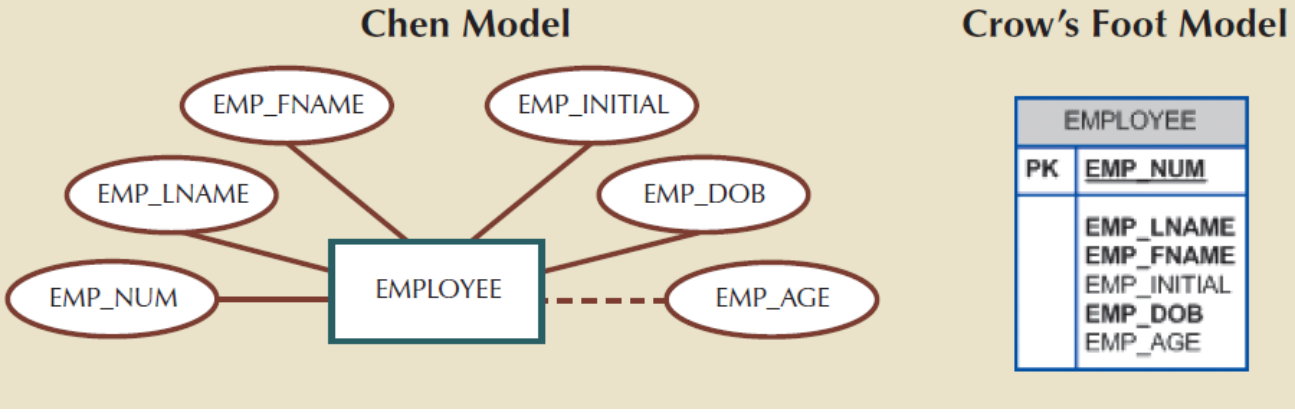
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Attributes (6 of 7)

Derived attribute: attribute whose value is calculated from other attributes

Derived using an algorithm

FIGURE 4.6 DEPICTION OF A DERIVED ATTRIBUTE



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Attributes (7 of 7)

Table 4.2	Advantages and Disadvantages of Storing Derived Attributes	
	Derived Attribute: Stored	**Derived Attribute: Not Stored**
Advantage	Saves CPU processing cycles Saves data access time Data value is readily available Can be used to keep track of historical data	Saves storage space Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	Uses CPU processing cycles Increases data access time Adds coding complexity to queries

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Derived Attribute – Example 1

```
CREATE TABLE COMPANYEMP (
C_EMP_CODE NUMBER(10) PRIMARY KEY,
C_EMP_TITLE VARCHAR2(4),
C_EMP_LNAME VARCHAR2(15) NOT NULL,
C_EMP_FNAME VARCHAR2(15) NOT NULL,
C_EMP_INITIAL VARCHAR2(1),
C_EMP_DOB DATE,
C_EMP_SALARY NUMBER(10) CHECK(C_EMP_SALARY > 0)
);
```

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flas

Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE
1	C_EMP_CODE	NUMBER(10,0)	No
2	C_EMP_TITLE	VARCHAR2(4 BYTE)	Yes
3	C_EMP_LNAME	VARCHAR2(15 BYTE)	No
4	C_EMP_FNAME	VARCHAR2(15 BYTE)	No
5	C_EMP_INITIAL	VARCHAR2(1 BYTE)	Yes
6	C_EMP_DOB	DATE	Yes
7	C_EMP_SALARY	NUMBER(10,0)	Yes

```
INSERT INTO COMPANYEMP VALUES (2201, 'Mr', 'Smith','Chippie',null, '22/01/1995',  65000.00);
INSERT INTO COMPANYEMP VALUES (2202, 'Ms', 'Zhao','Xin','Y', '31/03/1996', 68000.00);
INSERT INTO COMPANYEMP VALUES (2203, 'Mrs', 'Lin','Dan','B', '01/03/2000', 78000.00);
```


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Derived Attribute – Example 1

SELECT * FROM COMPANYEMP;

C_EMP_CODE	C_EM	C_EMP_LNAME	C_EMP_FNAME	C	C_EMP_DO	C_EMP_SALARY
2201	Mr	Smith	Chippie		22/01/95	65000
2202	Ms	Zhao	Xin	Y	31/03/96	68000
2203	Mrs	Lin	Dan	B	01/03/00	78000

SELECT C_EMP_TITLE, C_EMP_LNAME, C_EMP_FNAME, (SYSDATE - C_EMP_DOB)/365 C_EMP_AGE FROM COMPANYEMP;

C_EM	C_EMP_LNAME	C_EMP_FNAME	C_EMP_AGE
Mr	Smith	Chippie	27.5436669
Ms	Zhao	Xin	26.3546258
Mrs	Lin	Dan	22.4340778

SELECT C_EMP_TITLE, C_EMP_LNAME, C_EMP_FNAME, (ROUND((SYSDATE - C_EMP_DOB)/365,0)) C_EMP_AGE FROM COMPANYEMP;

C_EM	C_EMP_LNAME	C_EMP_FNAME	C_EMP_AGE
Mr	Smith	Chippie	28
Ms	Zhao	Xin	26
Mrs	Lin	Dan	22

DROP TABLE COMPANYEMP;

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Derived Attribute – Example 2

```
CREATE TABLE COMP_EMP (
CO_EMP_CODE NUMBER(10) PRIMARY KEY,  CO_EMP_TITLE VARCHAR2(4),
CO_EMP_LNAME VARCHAR2(15) NOT NULL, CO_EMP_FNAME VARCHAR2(15) NOT NULL,
CO_EMP_INITIAL VARCHAR2(1), CO_EMP_DOB DATE, CO_EMP_CITIZENSHIP VARCHAR2(15) NOT NULL,
CO_EMP_SALARY NUMBER(10) CHECK(CO_EMP_SALARY > 0),
CO_EMP_TYPE VARCHAR (6) GENERATED ALWAYS AS (
        CASE WHEN CO_EMP_CITIZENSHIP = 'New Zealand'
        THEN 'Local'
        ELSE 'OTHER'
        END)        VIRTUAL);
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT
1	CO_EMP_CODE	NUMBER(10,0)	No	(null)
2	CO_EMP_TITLE	VARCHAR2(4 BYTE)	Yes	(null)
3	CO_EMP_LNAME	VARCHAR2(15 BYTE)	No	(null)
4	CO_EMP_FNAME	VARCHAR2(15 BYTE)	No	(null)
5	CO_EMP_INITIAL	VARCHAR2(1 BYTE)	Yes	(null)
6	CO_EMP_DOB	DATE	Yes	(null)
7	CO_EMP_CITIZENSHIP	VARCHAR2(15 BYTE)	No	(null)
8	CO_EMP_SALARY	NUMBER(10,0)	Yes	(null)
9	CO_EMP_TYPE	VARCHAR2(6 BYTE)	Yes	CASE "CO_EMP_CITIZENSHIP" WHEN 'New Zealand' THEN 'Local' ELSE 'OTHER' END

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Derived Attribute – Example 2

INSERT INTO COMP_EMP (CO_EMP_CODE,CO_EMP_TITLE,CO_EMP_LNAME,CO_EMP_FNAME, CO_EMP_INITIAL, CO_EMP_DOB, CO_EMP_CITIZENSHIP, CO_EMP_SALARY)

VALUES (2201,'Mr', 'Smithie', 'Chippie', null, '22/01/1995', 'New Zealand', 65000.00);

INSERT INTO COMP_EMP (CO_EMP_CODE,CO_EMP_TITLE,CO_EMP_LNAME,CO_EMP_FNAME, CO_EMP_INITIAL, CO_EMP_DOB, CO_EMP_CITIZENSHIP, CO_EMP_SALARY)

VALUES(2202, 'Ms', 'Zhao', 'Xin', 'Y', '31/03/1996', 'Australia', 68000.00);

INSERT INTO COMP_EMP (CO_EMP_CODE, CO_EMP_TITLE, CO_EMP_LNAME, CO_EMP_FNAME, CO_EMP_INITIAL, CO_EMP_DOB, CO_EMP_CITIZENSHIP, CO_EMP_SALARY)

VALUES (2203, 'Ms', 'Zhao', 'Yin', 'M', '1/03/2000', 'Australia', 78000.00);

SELECT * FROM COMP_EMP;

	CO_EMP_CODE	CO_EMP_TITLE	CO_EMP_LNAME	CO_EMP_FNAME	CO_EMP_INITIAL	CO_EMP_DOB	CO_EMP_CITIZENSHIP	CO_EMP_SALARY	CO_EMP_TYPE
1	2201	Mr	Smithie	Chippie	(null)	22/01/95	New Zealand	65000	Local
2	2202	Ms	Zhao	Xin	Y	31/03/96	Australia	68000	OTHER
3	2203	Ms	Zhao	Yin	M	01/03/00	Australia	78000	OTHER

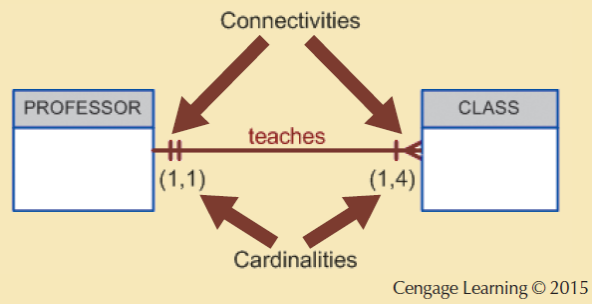
DROP TABLE COMP_EMP;

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Relationships, Connectivity, and Cardinality

- Association between entities that always operate in both directions
  - Participants: entities that participate in a relationship
- Connectivity: describes the relationship classification
  - Include 1:1, 1:M, and M:N
- Cardinality: expresses the minimum and maximum number of entity occurrences associated with one occurrence of related entity
  - In the ERD, cardinality is indicated by placing the appropriate numbers beside the entities, using the format (x, y)



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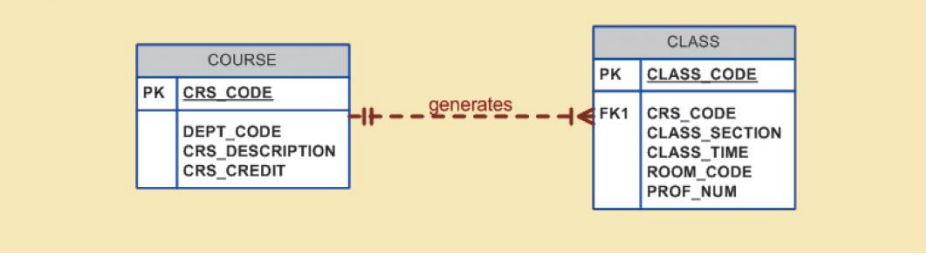
Existence Dependence

- Existence dependence
  - Entity exists in the database only when it is associated with another related entity occurrence
- Existence independence
  - Entity exists apart from all of its related entities
  - Referred to as a strong entity or regular entity

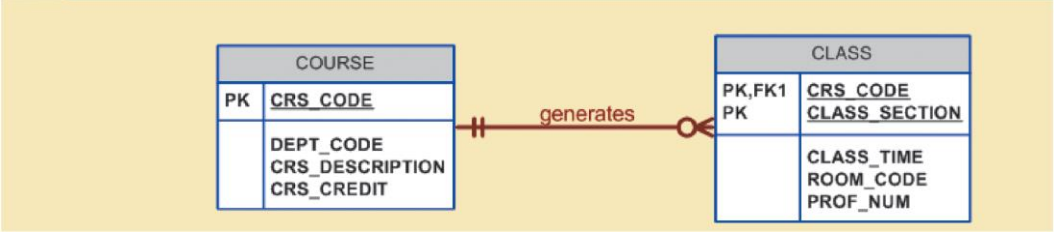
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Relationship Strength

- Weak (non-identifying) relationship
  - Primary key of the related entity does not contain a primary key component of the parent entity



- Strong (identifying) relationships
  - Primary key of the related entity contains a primary key component of the parent entity



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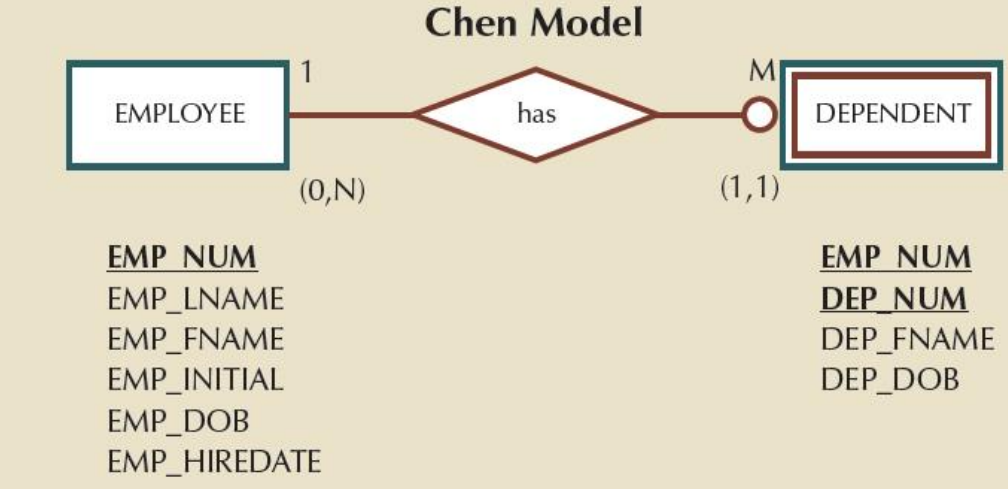
Weak Entities (1 of 3)

- Conditions of a weak entity
  - Existence-dependent
  - Has a primary key that is partially or totally derived from parent entity in the relationship
- Database designer determines whether an entity is weak
  - Based on business rules

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Weak Entities (2 of 3)

FIGURE 4.10 A WEAK ENTITY IN AN ERD



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Weak Entities (3 of 3)

FIGURE 4.11 A WEAK ENTITY IN A STRONG RELATIONSHIP

Table name: EMPLOYEE **Database name: Ch04_ShortCo**

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
1001	Callifante	Jeanine	J	12-Mar-64	25-May-97
1002	Smithson	William	K	23-Nov-70	28-May-97
1003	Washington	Herman	H	15-Aug-68	28-May-97
1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
1005	Johnson	Melanie		28-Sep-66	20-Dec-98
1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

Table name: DEPENDENT

EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
1001	1	Annelise	05-Dec-97
1001	2	Jorge	30-Sep-02
1003	1	Suzanne	25-Jan-04
1006	1	Carlos	25-May-01
1008	1	Michael	19-Feb-95
1008	2	George	27-Jun-98
1008	3	Katherine	18-Aug-03

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Relationship Participation (1 of 3)

- Optional participation
  - One entity occurrence does not require a corresponding entity occurrence in a particular relationship
- Mandatory participation
  - One entity occurrence requires a corresponding entity occurrence in a particular relationship

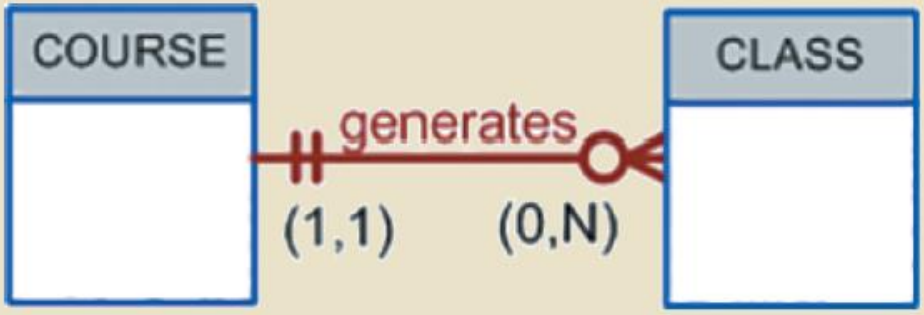
CROW'S FOOT SYMBOLS	CARDINALITY	COMMENT
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

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Relationship Participation (2 of 3)

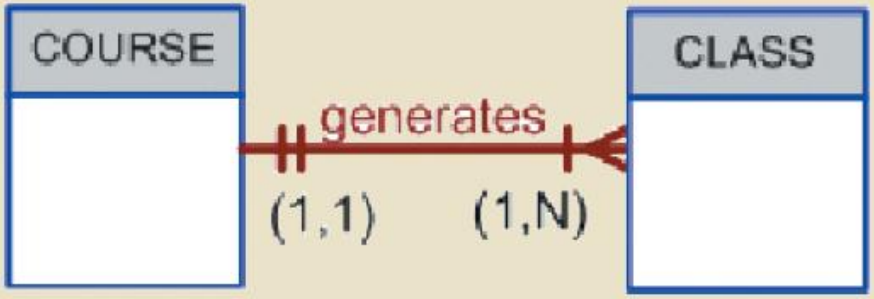
FIGURE 4.13 CLASS IS OPTIONAL TO COURSE



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Relationship Participation (3 of 3)

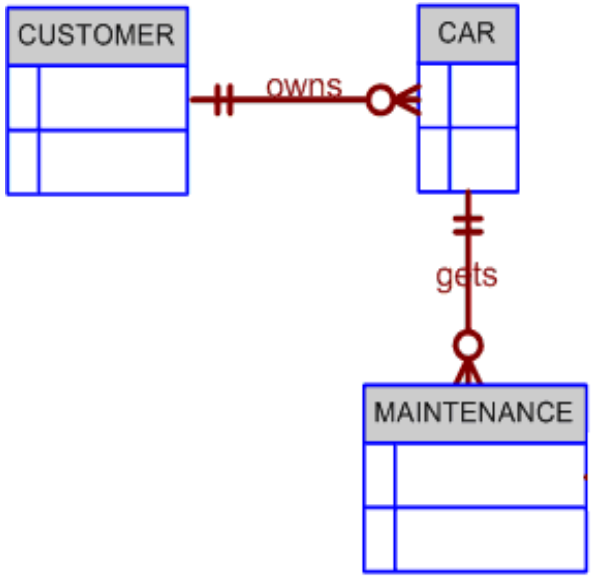
FIGURE 4.14 COURSE AND CLASS IN A MANDATORY RELATIONSHIP



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Business Rules

1. A customer may own many cars.
2. A car is owned by only one customer.
3. A car may generate one or more maintenance records
4. Each maintenance record is generated by one and only one car.



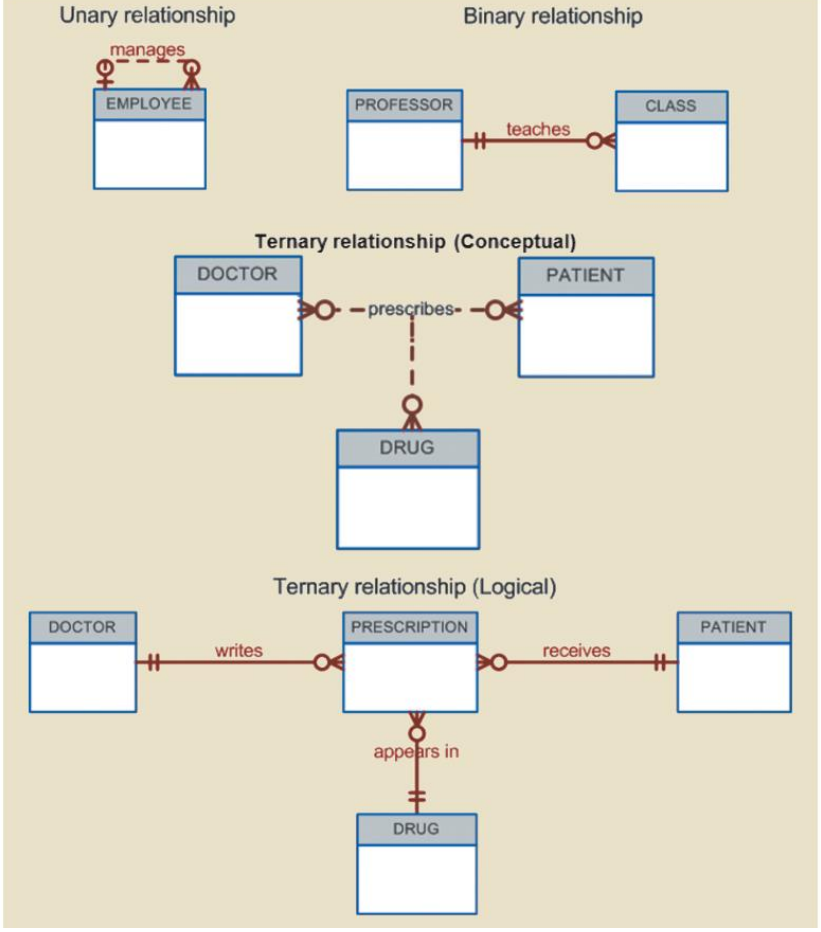
Relationship Degree (1 of 2)

- Indicates the number of entities or participants associated with a relationship
  - Unary relationship: association is maintained within a single entity
  - Binary relationship: two entities are associated
  - Ternary relationship: three entities are associated
  - Recursive relationship: relationship exists within a single entity type

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Relationship Degree (2 of 2)

FIGURE 4.15 THREE TYPES OF RELATIONSHIP DEGREE



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Recursive Relationships (1 of 2)

- Relationship can exist between occurrences of the same entity set
  - Naturally, such a condition is found within a unary relationship
    - Common in manufacturing industries
- One common pitfall when working with unary relationships is to confuse participation with referential integrity
  - Similar because they are both implemented through constraints on the same set of attributes

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Recursive Relationships (2 of 2)

FIGURE 4.17 AN ER REPRESENTATION OF RECURSIVE RELATIONSHIPS



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Unary relationship



Table name: EMPLOYEE_V2

EMP_CODE	EMP_LNAME	EMP_MANAGER
101	Waddell	102
102	Orincona	
103	Jones	102
104	Reballoh	102
105	Robertson	102
106	Deltona	102

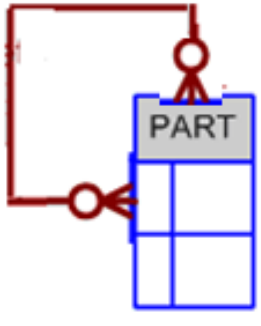


Table name: COMPONENT

COMP_CODE	PART_CODE	COMP_PARTS_NEEDED
C-130	AA21-6	4
C-130	AB-121	2
C-130	E129	1
C-131A2	E129	1
C-130	X10	4
C-131A2	X10	1
C-130	X34AW	2
C-131A2	X34AW	2

Table name: PART

PART_CODE	PART_DESCRIPTION	PART_IN_STOCK
AA21-6	2.5 cm. washer, 1.0 mm. rim	432
AB-121	Cotter pin, copper	1034
C-130	Rotor assembly	36
E129	2.5 cm. steel shank	128
X10	10.25 cm. rotor blade	345
X34AW	2.5 cm. hex nut	879

Database Design Challenges: Conflicting Goals (1 of 2)

- Database designers must often make design compromises that are triggered by conflicting goals
  - Database design must conform to design standards
  - High processing speed may limit the number and complexity of logically desirable relationships
  - Maximum information generation may lead to loss of clean design structures and high transaction speed

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Database Design Challenges: Conflicting Goals (2 of 2)

FIGURE 4.38 VARIOUS IMPLEMENTATIONS OF THE 1:1 RECURSIVE RELATIONSHIP

Table name: EMPLOYEE_V1 — **Database name: Ch04_PartCo**

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_SPOUSE
345	Ramirez	James	347
346	Jones	Anne	349
347	Ramirez	Louise	345
348	Delaney	Robert	
349	Shapiro	Anton	346

First implementation

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME
345	Ramirez	James
346	Jones	Anne
347	Ramirez	Louise
348	Delaney	Robert
349	Shapiro	Anton

Table name: MARRIED_V1

EMP_NUM	EMP_SPOUSE
345	347
346	349
347	345
349	346

Second implementation

Table name: MARRIAGE

MAR_NUM	MAR_DATE
1	04-Mar-03
2	02-Feb-99

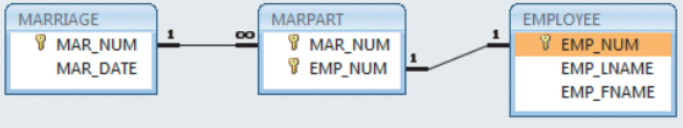
Table name: MARPART

MAR_NUM	EMP_NUM
1	345
1	347
2	346
2	349

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME
345	Ramirez	James
346	Jones	Anne
347	Ramirez	Louise
348	Delaney	Robert
349	Shapiro	Anton

The relational diagram for the third implementation



Third implementation

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Associative (Composite) Entities (1 of 2)

- Used to represent an M:N relationship between two or more entities
- Has a 1:M relationship with the parent entities
  - Composed of the primary key attributes of each parent entity
- May also contain additional attributes that play no role in connective process

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Associative (Composite) Entities (2 of 2)

FIGURE 4.25 A COMPOSITE ENTITY IN AN ERD

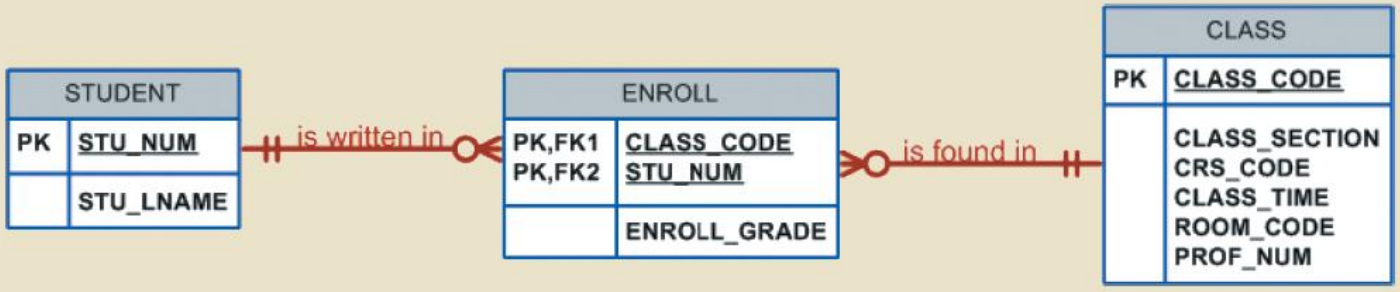


Table name: STUDENT

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Table name: ENROLL

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114

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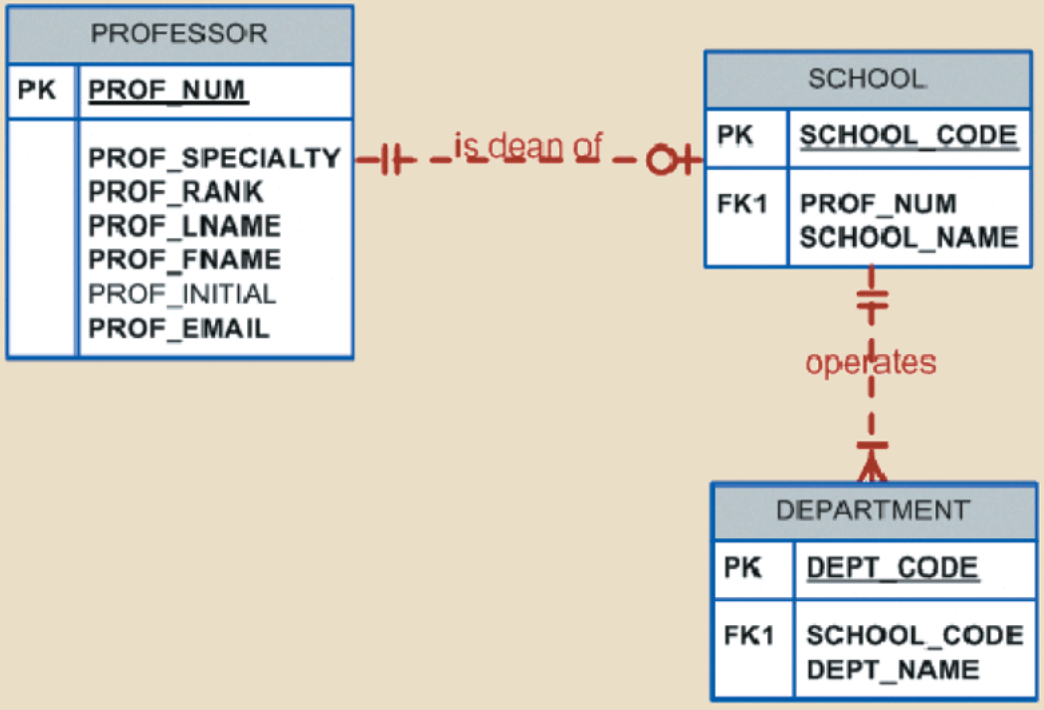
Developing an ER Diagram (1 of 11)

- Activities involved in building and ERD
  - Create a detailed narrative of the organization's description of operations
  - Identify business rules based on the descriptions
  - Identify main entities and relationships from the business rules
  - Develop the initial ERD
  - Identify the attributes and primary keys that adequately describe entities
  - Revise and review ERD

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Developing an ER Diagram (2 of 11)

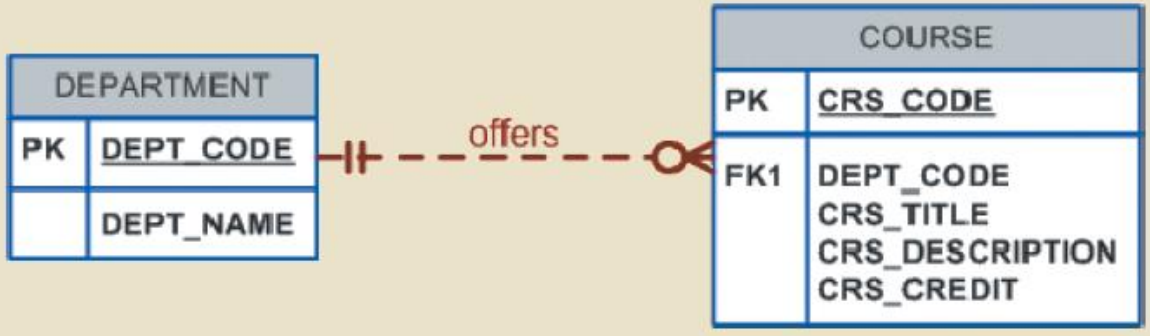
FIGURE 4.26 THE FIRST TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (3 of 11)

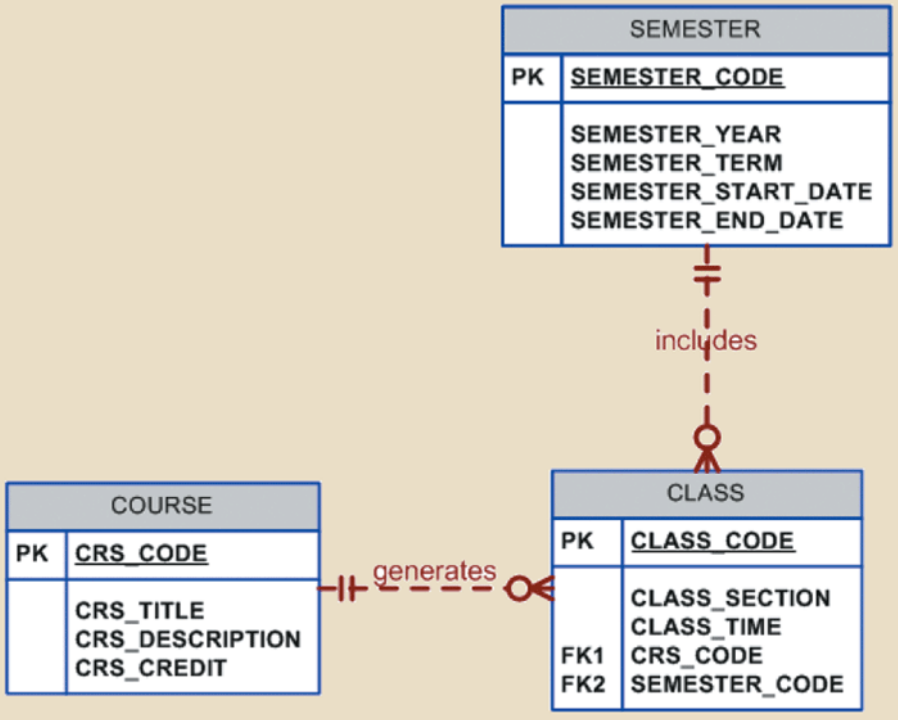
FIGURE 4.27 THE SECOND TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (4 of 11)

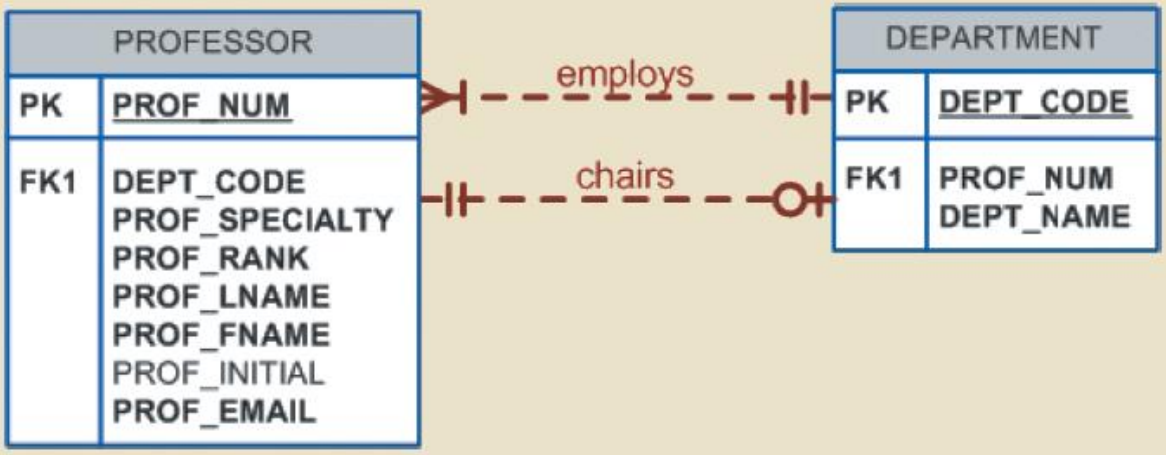
FIGURE 4.28 THE THIRD TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (5 of 11)

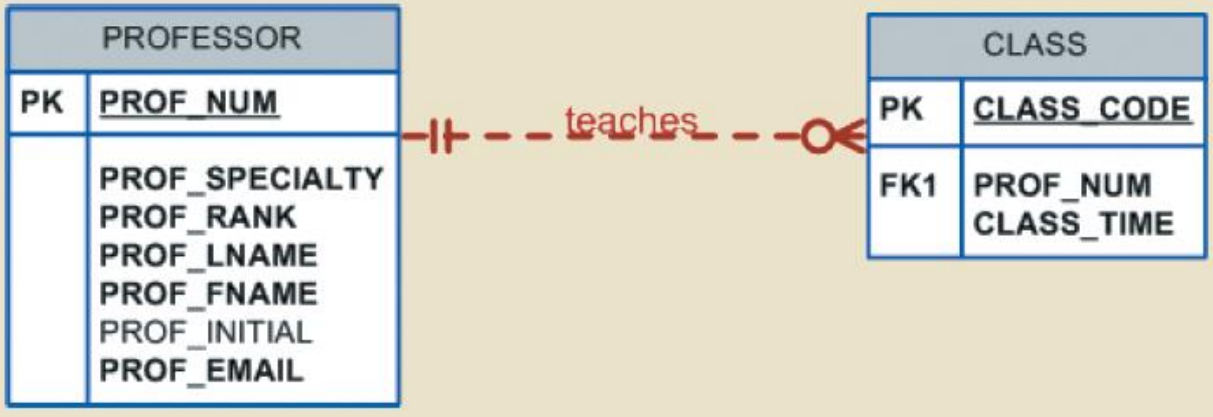
FIGURE 4.29 THE FOURTH TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (6 of 11)

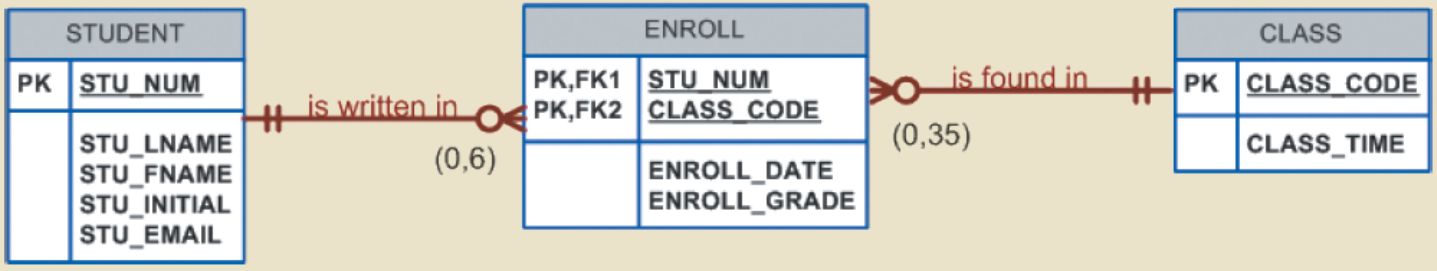
FIGURE 4.30 THE FIFTH TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (7 of 11)

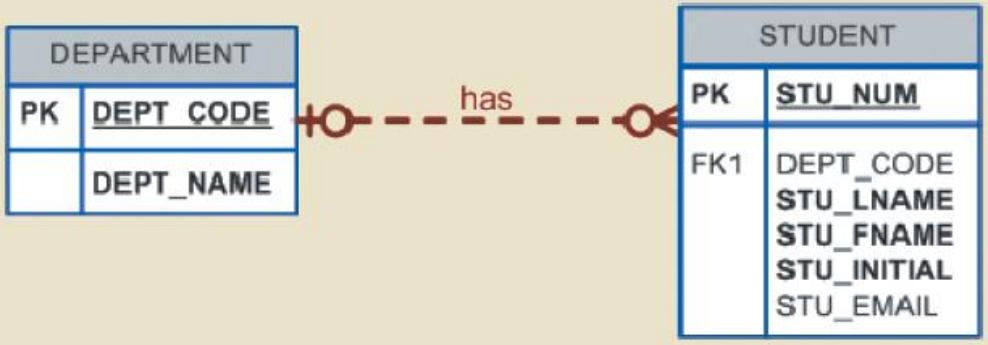
FIGURE 4.31 THE SIXTH TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (8 of 11)

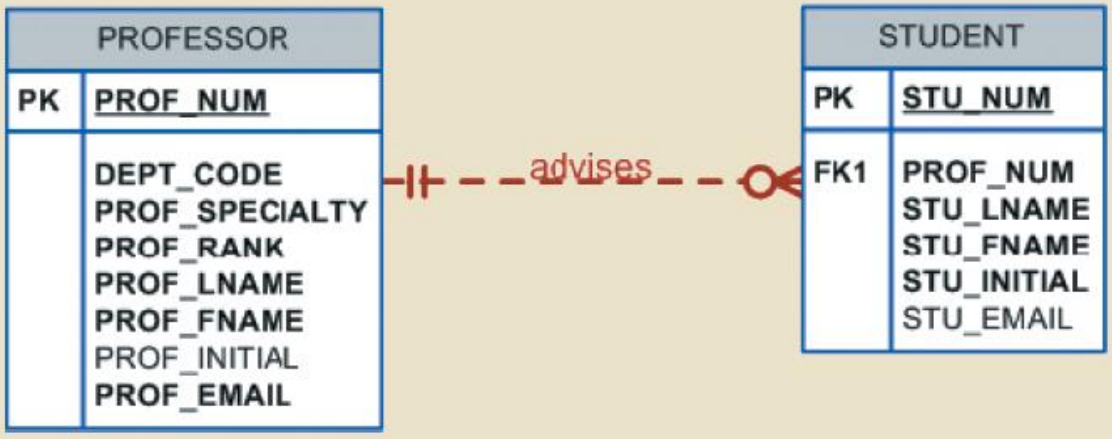
FIGURE 4.32 THE SEVENTH TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (9 of 11)

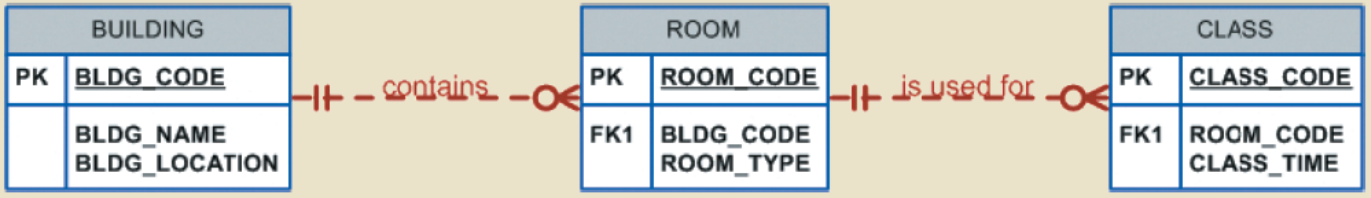
FIGURE 4.33 THE EIGHT TINY COLLEGE ERD SEGMENT



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Developing an ER Diagram (10 of 11)

FIGURE 4.34 THE NINTH TINY COLLEGE ERD SEGMENT



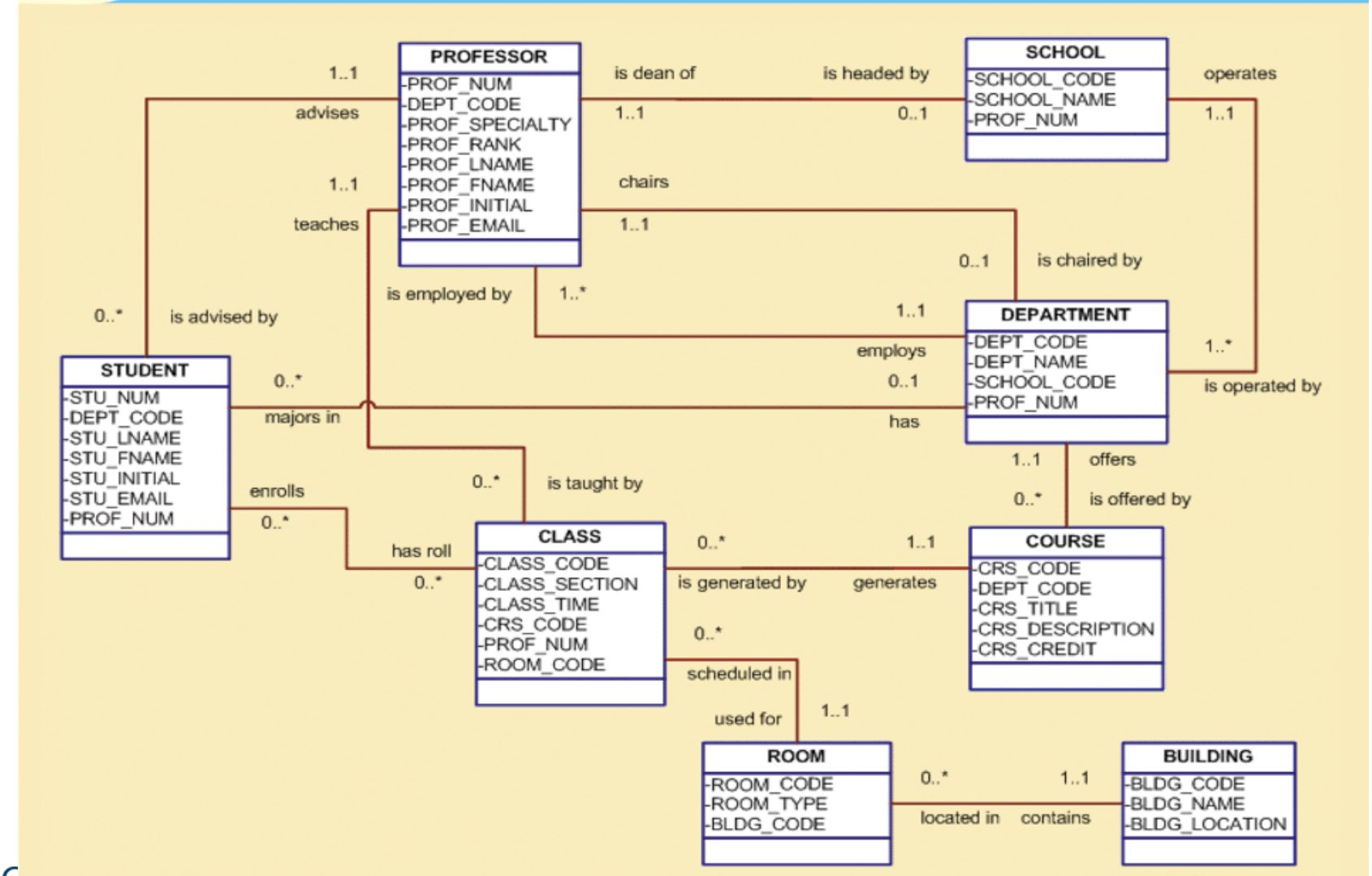
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Developing an ER Diagram (11 of 11)

Table 4.4	Components of the ERM		
Entity	**Relationship**	**Connectivity**	**Entity**
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
SEMESTER	includes	1:M	CLASS
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	M:N	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS
	Note: ENROLL is the composite entity that implements the M:N relationship “STUDENT enrolls in CLASS.”		

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FIGURE 4.36 The conceptual UML class diagram for Tiny College



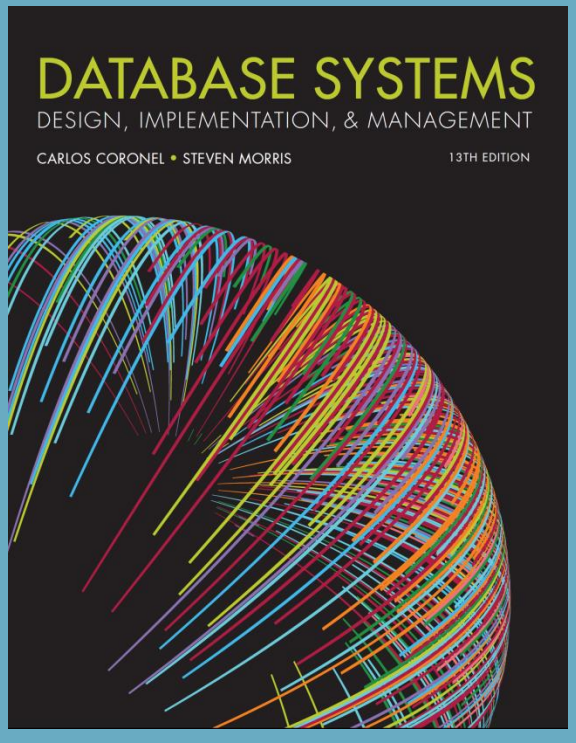
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Summary

- The ERM uses ERDs to represent the conceptual database as viewed by the end user
- Connectivity describes the relationship classification (1:1, 1:M, or M:N)
- In the ERM, an M:N relationship is valid at the conceptual level
- ERDs may be based on many different ERMs
- Unified Modeling Language (UML) class diagrams are used to represent the static data structures in a data model
- Database designers, no matter how well they can produce designs that conform to all applicable modeling conventions, are often forced to make design compromises

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Chapter 5 (only section 5.3 and 5.4)

Advanced Data Modeling

Entity Integrity: Selecting Primary Keys

- Primary keys: single attribute or a combination of attributes
  - Uniquely identifies each entity instance
  - Guarantees entity integrity
  - Works with foreign keys to implement relationships

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Natural Keys and Primary Keys

- Natural key or natural identifier: real-world identifier used to uniquely identify real-world objects
  - Familiar to end users and forms part of their day-to-day business vocabulary
  - Used as the primary key of the entity being modeled

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Primary Key Guidelines

- Desirable primary key characteristics
  - Non intelligent
  - No change over time
  - Preferably single-attribute
  - Preferably numeric
  - Security-compliant

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When to Use Composite Primary Keys (1 of 2)

- Identifiers of composite entities
  - Each primary key combination is allowed once in M:N relationship
- Identifiers of weak entities
  - Strong identifying relationship with the parent entity
  - Represents a real-world object that is existence-dependent on another real-world object
  - Represented in the data model as two separate entities in a strong identifying relationship

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When to Use Composite Primary Keys (2 of 2)

FIGURE 5.7 THE M:N RELATIONSHIP BETWEEN STUDENT AND CLASS

Database name: Ch05_Tinycollege

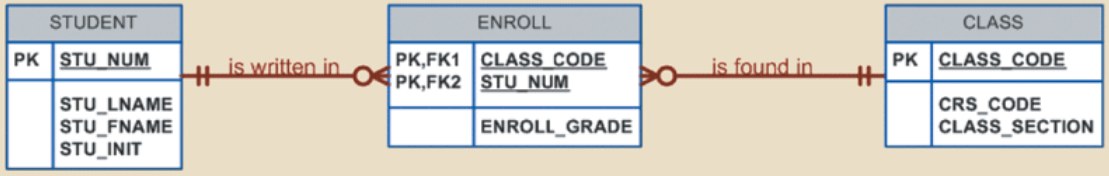


Table name: STUDENT
(first four fields)

STU_NUM	STU_LNAME	STU_FNAME	STU_INIT
321452	Bowser	William	C
324257	Smithson	Anne	K
324258	Brewer	Juliette	
324269	Oblonski	Walter	H
324273	Smith	John	D
324274	Katinga	Raphael	P
324291	Robertson	Gerald	T
324299	Smith	John	B

Table name: ENROLL

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS
(first three fields)

CLASS_CODE	CRS_CODE	CLASS_SECTION
10012	ACCT-211	1
10013	ACCT-211	2
10014	ACCT-211	3
10015	ACCT-212	1
10016	ACCT-212	2
10017	CIS-220	1
10018	CIS-220	2
10019	CIS-220	3
10020	CIS-420	1
10021	QM-261	1
10022	QM-261	2
10023	QM-362	1
10024	QM-362	2
10025	MATH-243	1

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When to Use Surrogate Primary Keys (1 of 2)

- Primary key used to simplify the identification of entity instances
  - Useful when there is no natural key
  - Helpful if selected candidate key has embedded semantic contents or is too long
- Require ensuring that the candidate key of entity in question performs properly
  - Use “unique index” and “not null” constraints

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When to Use Surrogate Primary Keys (2 of 2)

Table 5.4: Data Used to Keep Track of Events					
DATE	TIME_START	TIME_END	ROOM	EVENT_NAME	PARTY_OF
6/17/2018	11:00 a.m.	2:00 p.m.	Allure	Burton Wedding	60
6/17/2018	11:00 a.m.	2:00 p.m.	Bonanza	Adams Office	12
6/17/2018	3:00 p.m.	5:30 p.m.	Allure	Smith Family	15
6/17/2018	3:30 p.m.	5:30 p.m.	Bonanza	Adams Office	12
6/18/2018	1:00 p.m.	3:00 p.m.	Bonanza	Boy Scouts	33
6/18/2018	11:00 a.m.	2:00 p.m.	Allure	March of Dimes	25
6/18/2018	11:00 a.m.	12:30 p.m.	Bonanza	Smith Family	12

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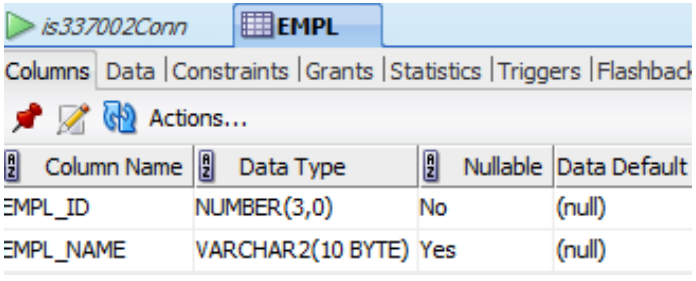
Using Sequences to Generate Key Values

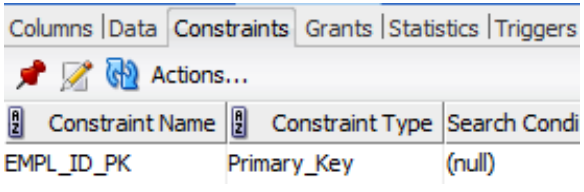
- Generally, during implementation, we generate unique values for each record added through use of SQL code or application software, thus eliminating the need for data entry by humans.

- Sequences – allow you to generate unique integers

```
EMPL(EMPL_ID NUMBER(3), EMPL_NAME VARCHAR2(10))

PROMPT "CREATE TABLE EMPL USING SEQUENCES"
CREATE TABLE EMPL(EMPL_ID NUMBER(3)  CONSTRAINT EMPL_ID_PK PRIMARY KEY,
   EMPL_NAME VARCHAR2(10));
```





Using Sequences

```
INSERT INTO EMPL VALUES(1, 'CARLOS');
INSERT INTO EMPL VALUES(2, 'STEVEN ');
INSERT INTO EMPL VALUES(3, 'PETER ');
```

is337002Conn | EMPL

Columns | Data | Constraints | Grants | Stat

Sort... | Fil

	EMPL_ID	EMPL_NAME
1	1	CARLOS
2	2	STEVEN
3	3	PETER

```
CREATE SEQUENCE SEQ_EMPL_ID
START WITH 4
NOCACHE
NOCYCLE;
```



is337002Conn | SEQ_EMPL_ID

Details | Dependencies | SQL

Actions...

Name	Value
CREATED	16/03/12
LAST_DDL_TIME	16/03/12
SEQUENCE_OWNER	IS337002
SEQUENCE_NAME	SEQ_EMPL_ID
MIN_VALUE	1
MAX_VALUE	999999999999999999999999999
INCREMENT_BY	1
CYCLE_FLAG	N
ORDER_FLAG	N
CACHE_SIZE	0
LAST_NUMBER	4

Using Sequences

```
INSERT INTO EMPL VALUES (SEQ_EMPL_ID.NEXTVAL, 'SAMUEL');
INSERT INTO EMPL VALUES (SEQ_EMPL_ID.NEXTVAL, 'JANE');

DROP SEQUENCE SEQ_EMPL_ID;
DROP TABLE EMPL;
```

is337002Conn | EMPL

Columns | Data | Constraints | Grants | Sta

Sort... | F

	EMPL_ID	EMPL_NAME
1	1	CARLOS
2	2	STEVEN
3	3	PETER
4	4	SAMUEL
5	5	JANE

Design Case 1: Implementing 1:1 Relationships (1 of 3)

- Foreign keys work with primary keys to properly implement relationships in relational model
  - Place primary key of the parent entity on the dependent entity as foreign key
- Options for selecting and placing the foreign key
  - Place a foreign key in both entities
  - Place a foreign key in one of the entities

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Design Case 1: Implementing 1:1 Relationships (2 of 3)

Table 5.5: Selection of Foreign Key in a 1:1 Relationship		
Case	**ER Relationship Constraints**	**Action**
I	One side is mandatory and the other side	Place the PK of the entity on the mandatory side in the entity on the optional side as a FK, and make the FK mandatory
II	Both sides are optional	Select the FK that causes the fewest nulls, or place the FK in the entity in which the (relationship) role is played
III	Both sides are mandatory	See Case II, or consider revising your model to ensure that the two entities do not belong together in a single entity

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Design Case 1: Implementing 1:1 Relationships (3 of 3)

FIGURE 5.8 THE 1:1 RELATIONSHIP BETWEEN DEPARTMENT AND EMPLOYEE

A One-to-One (1:1) Relationship:
An EMPLOYEE manages zero or one DEPARTMENT;
each DEPARTMENT is managed by one EMPLOYEE.



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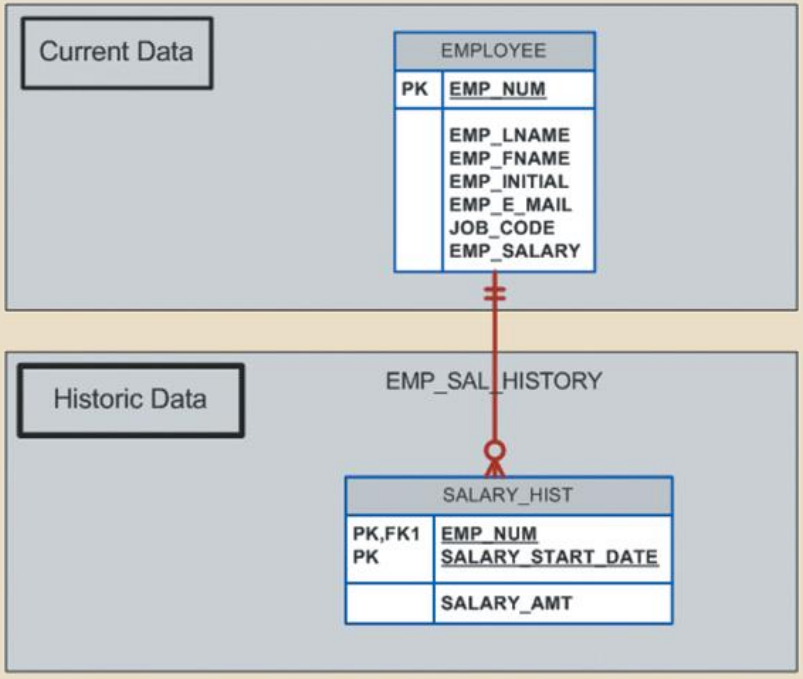
Design Case 2: Maintaining History of Time-Variant Data (1 of 4)

- Time-variant data: data whose values change over time and for which a history of the data changes must be retained
  - Requires creating a new entity in a 1:M relationship with the original entity
  - New entity contains the new value, date of the change, and any other pertinent attribute

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Design Case 2: Maintaining History of Time-Variant Data (2 of 4)

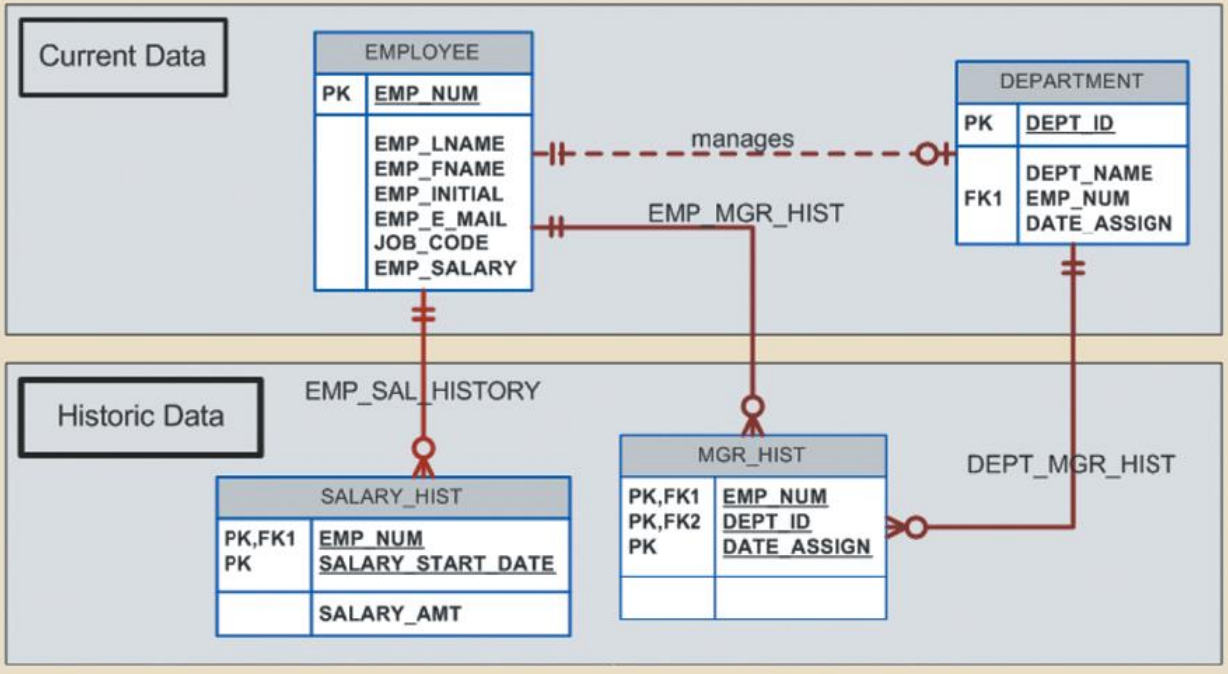
FIGURE 5.9 MAINTAINING SALARY HISTORY



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Design Case 2: Maintaining History of Time-Variant Data (3 of 4)

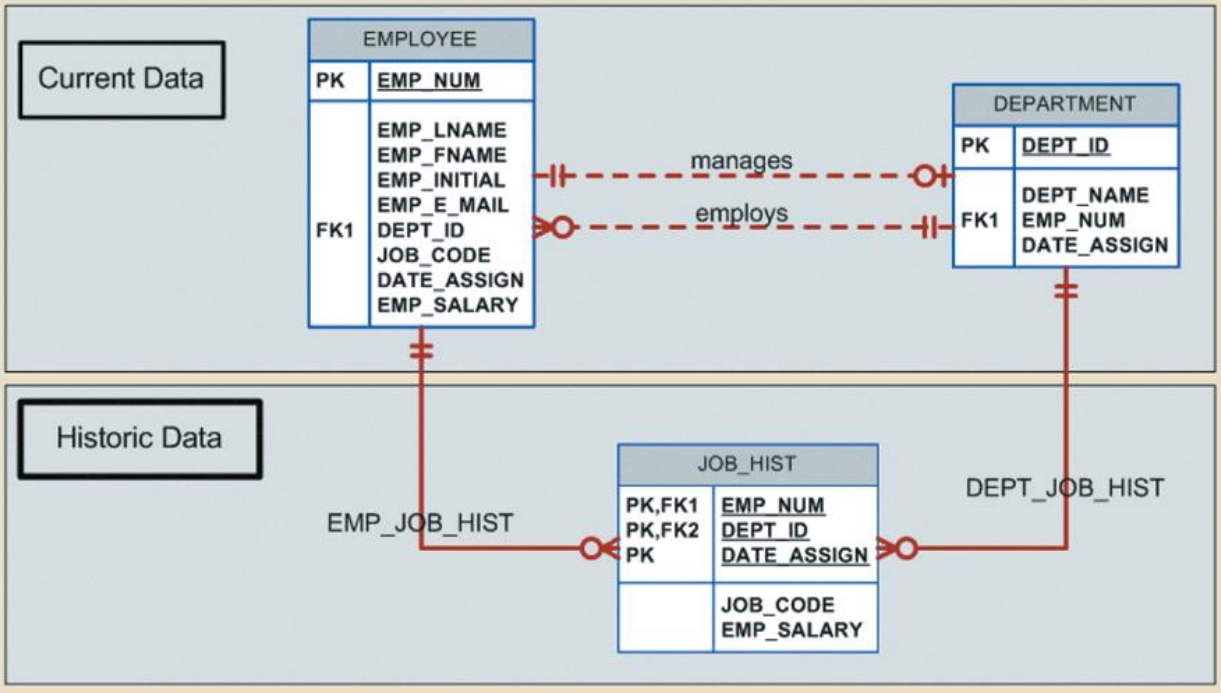
FIGURE 5.10 MAINTAINING MANAGER HISTORY



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Design Case 2: Maintaining History of Time-Variant Data (4 of 4)

FIGURE 5.11 MAINTAINING JOB HISTORY



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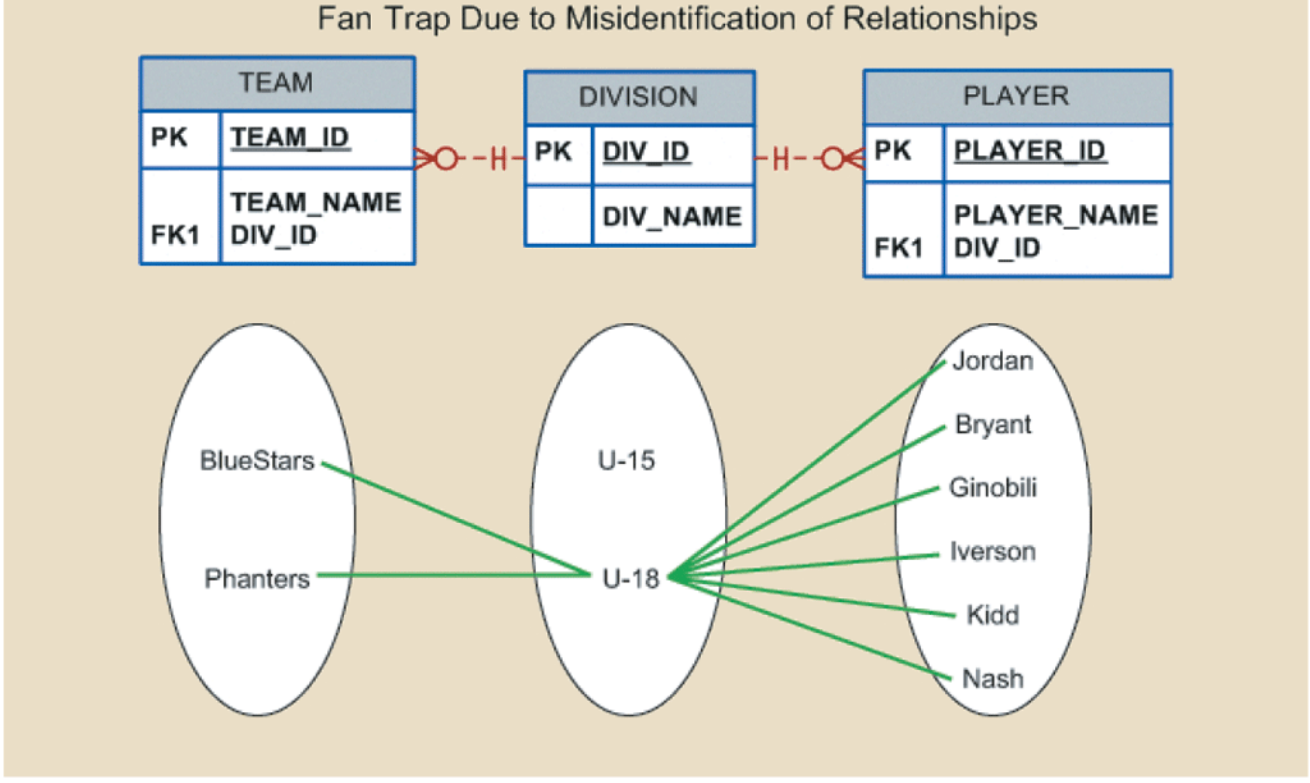
Design Case 3: Fan Traps (1 of 3)

- Design trap: occurs when a relationship is improperly or incompletely identified
  - Represented in a way not consistent with the real world
- Fan trap: occurs when one entity is in two 1:M relationships to other entities
  - Produces an association among other entities not expressed in the model

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Design Case 3: Fan Traps (2 of 3)

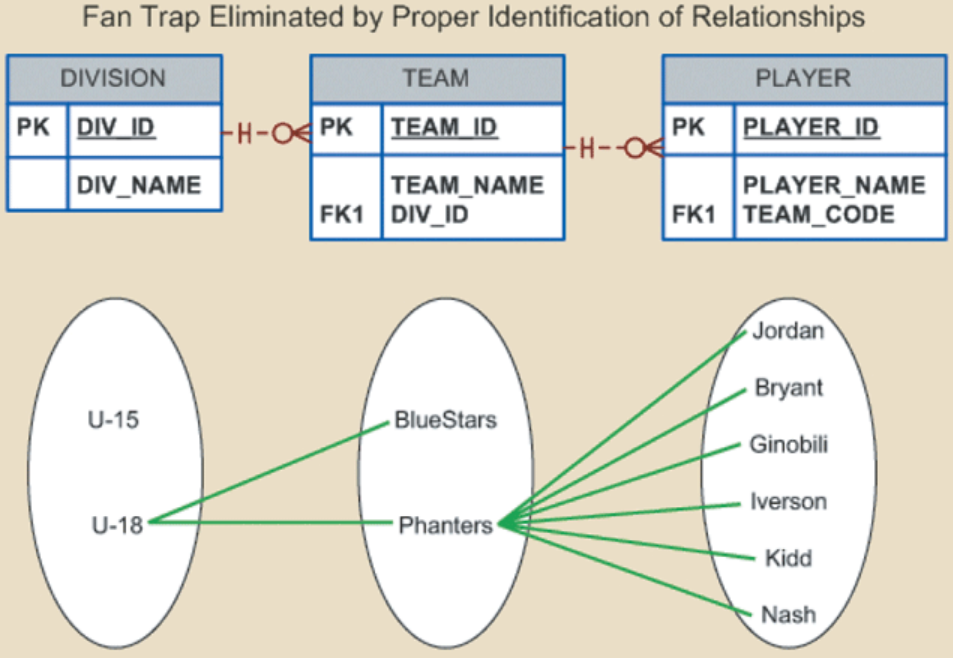
FIGURE 5.12 INCORRECT ERD WITH FAN TRAP PROBLEM



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Design Case 3: Fan Traps (3 of 3)

FIGURE 5.13 CORRECTED ERD AFTER REMOVAL OF THE FAN TRAP

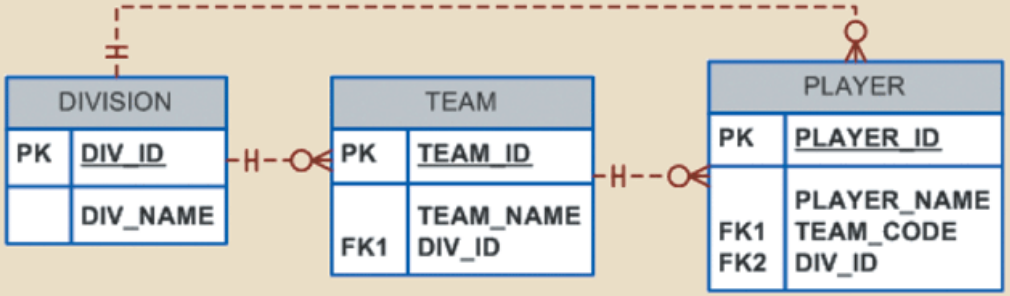


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Design Case 4: Redundant Relationships

- Occur when there are multiple relationship paths between related entities
  - Must remain consistent across the model
  - Help simplify the design

FIGURE 5.14 A REDUNDANT RELATIONSHIP



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